Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I
Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 08.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 3

Topics to be discussed: The Electric Field
Definition of electric field, derivation an equation for the electric field, electric lines of force, electric field for a system of charge, solution of some mathematical problems.

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define and calculate the electric field
- ❖ Explain the concept of electric lines of force
- ❖ Explain the procedure to calculate electric field strength at a location in the vicinity of two opposite charges with a numerical example

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

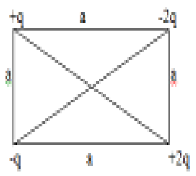
Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) What is the unit of the electric field?
   a) N b) C c) F d) NC^{-1} e) CN^{-1}
2) Tangent at any point in lines of force indicated the –
   a) potential. b) dielectric constant. c) direction of electric field.
   d) electric force. e) electric momentum.

Sample Questions (SAQ):

1) Define electric field. How do you determine the electric field strength of a unit charge?
2) Define field line and explain its purpose. Demonstrate repulsion and attraction by line of force.
3) Find the electric filed due to a long straight uniform charged wire.
4) Find out the value of electric filed at a point on the axis of a charged circular turn of wire.
5) Two charges $q_1 = 1 \times 10^{-6} C$ and $q_2 = 2 \times 10^{-6} C$ are kept 10 cm away from each other. At what point the line joining the two charges is the electric field strength zero?
6) What is the magnitude of the electric field strength E such that an electron, placed in the field would experience an electrical force equal to its weight? (Mass of the electron = $9.1 \times 10^{-31} kg$)
7) What is the electric field E in magnitude at the centre of the following square? Assume that, $q = 1 \times 10^{-8} C$ and $a = 5.0 cm$.



Reference Books:
1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 326-331
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86